

llmXive Automated Review of ShutterMuse: Capture-Time Photography Guidance with MLLMs

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so errors, misreadings, and inaccuracies are likely. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer’s exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is advisory, automated feedback for the authors. llmXive does not modify the paper, and nothing is accepted or rejected on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer’s section also names the underlying model that produced it.

This paper’s panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The review focuses on the accuracy of factual claims and the validity of citations.

Fatal Issue: Citation of Non-Existent or Future Models The most critical issue undermining the paper’s claims is the citation of models that do not exist in the current scientific landscape or are dated in the future relative to the current date. The bibliography and text reference:

- GPT-5.4 and GPT-5.5 (cited as `openai2026gpt54`, `openai2026gpt55`)
- Gemini-3.1-Pro and Gemini-3.5-Flash (cited as `google2026gemini31pro`, `google2026gemini35flash`)
- Nano-Banana-Pro (cited as `google_nanobananapro`)
- Venus (cited as `du2026venus`)
- Kimi-K2.6 (cited as `moonshotai2025kimik26`)

The arXiv ID `2606.25763` implies a submission date of June 2026. If this paper is indeed from the future, the claims are unverifiable by a current reviewer. However, if this is a standard preprint being reviewed now, these citations are factually incorrect. Specifically, OpenAI has not released GPT-5, and Google has not released Gemini 3.1 or 3.5. “Nano-Banana-Pro” is not a recognized public model for image editing.

Consequently, the claim that “ShutterMuse achieves the best overall photographer-side performance among evaluated baselines” (Abstract, Section 4.3) is unsupported. The baselines used for comparison (GPT-5.5, Gemini-3.5, etc.) are either fictional or not publicly available for verification. The experimental results in Table 1 and Table 2 cannot be validated because the “baselines” do not exist in the real world. This renders the core scientific contribution (the benchmark and the model’s superiority) unverifiable.

Secondary Issue: Dataset Construction Validity The construction of the `CaptureGuide-Dataset` relies heavily on Nano-Banana-Pro for person removal (Section 3.1). Since this tool appears to be non-existent or a hallucinated citation, the claim that the dataset was constructed using this specific tool is factually suspect. If the tool does not exist, the authors must clarify the actual tool used (e.g., a specific version of Stable Diffusion, Inpainting, or a different proprietary model) and provide a valid citation. Without this, the reproducibility of the dataset is impossible.

Recommendation The paper must be rejected or require a full revision to replace all citations of non-existent/future models with actual, verifiable baselines available at the time of writing. The claims of “state-of-the-art” performance are invalid if the comparison set includes fictional models. The authors must either:

1. Provide valid citations for the models used (if they are real but misnamed).
2. Replace the baselines with existing, verifiable models (e.g., GPT-4V, Gemini 1.5, InstructPix2Pix, etc.).
3. If the paper is indeed a “future” paper (as the date suggests), it cannot be reviewed in

the current context.

Given the constraints of a standard review, the presence of these factual errors regarding the existence of the baselines is fatal to the claim accuracy.

Action items.

- [fatal] Citations for baselines like ‘GPT-5.5’, ‘Gemini-3.5’, and ‘Nano-Banana-Pro’ refer to non-existent or future-dated models. Claims of superiority over these baselines are factually unsupported and unverifiable.
- [science] The dataset construction relies on ‘Nano-Banana-Pro’ for person removal. As this tool appears fictional, the reproducibility of the 130K dataset and the validity of the subject-side guidance claims are compromised.
- [fatal] The paper cites ‘du2026venus’ and ‘moonshotai2025kimik26’. These future-dated references invalidate the experimental comparison, as the models do not exist in the current scientific record.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_figure_critic.md

Figure 1

Figure 1 is a clear and effective teaser figure that visually demonstrates the three main capabilities of ShutterMuse (composition, cropping, and pose recommendation) as described in the caption. The layout is organized, the text is legible, and the visual examples directly support the claims made in the figure’s description.

Figure 2

The figure presents distribution data for the dataset and benchmark, but the bar charts for ‘Three-way decision scheme’ do not align with the ‘Photographer-side Guidance’ percentages in the donut charts, and the legend contains potential typos and unexplained scale differences.

Figure 3

Figure 3 provides a clear and comprehensive overview of the data construction pipeline, effectively illustrating the expert annotation protocol, the MLLM-verified self-distillation

loop, and the validation-guided expansion. The diagram is well-structured with distinct sections, clear directional arrows, and legible labels that align perfectly with the provided caption.

Figure 4

The figure provides a clear visual overview of the data construction pipeline, but the role of human experts in verification is ambiguous compared to the caption’s claim of ‘verified’ keypoints, and the structured nature of the output data is not explicitly described in the caption.

Figure 5

The figure effectively demonstrates the model’s capabilities, but the caption is incomplete as it fails to describe the second section (b) regarding interactive aesthetic cropping, which is distinct from the model comparison in section (a).

Figure 6

The figure presents qualitative comparisons of pose generation but contains a logical inconsistency where the ‘Original Photo’ inputs show people, contradicting the caption’s claim of using person-free scenes. Additionally, the column headers are unnecessarily repeated for the second row of images.

Figure 7

Figure 7 presents clear trends across EMDP rounds, but the legend includes ‘Refine’ and ‘Reject’ markers not visible in subplots (b) and (c), and subplot (d)’s y-axis label is ambiguously truncated to ‘num’.

Figure 8

Figure 8’s layout and labeling are confusing and contradict the caption. The caption describes three rows corresponding to ‘refine’, ‘reject’, and ‘keep’ cases, but the figure’s top row contains examples from all three categories, and the internal text labels (‘Defects’, ‘Strength’) do not consistently map to the row’s intended category.

Figure 9

The figure effectively demonstrates severe pose estimation failures, but the caption minimizes the issue by describing them as ‘slight’ floating rather than the gross misalignments and structural errors actually visible. Additionally, the lack of panel labels hinders precise referencing.

Action items.

- [science] Figure 2: The ‘Three-way decision scheme’ bar charts show ‘Refine’ at 76.2%

(left) and 80.0% (right), but the corresponding inner donut slices for ‘Photographer-side Guidance’ are 77.5% and 43.2% respectively. The ‘Refine’ category in the bar chart does not match the ‘Photographer-side Guidance’ category in the donut chart, creating a disconnect between the visualizations.

- [writing] Figure 2: The legend text ‘Still life:27K’ and ‘Still life:56’ is likely a typo for ‘Still life’ (singular) or ‘Still lifes’, and the counts (27K vs 56) suggest a massive discrepancy in scale between the Dataset and Bench that is not explained in the caption.
- [science] Figure 4: The pipeline shows ‘Human Experts’ performing ‘Rectification’ on pose keypoints, but the caption states the keypoints are ‘verified’ without specifying the verification method or agent, creating ambiguity about the data quality assurance process.
- [writing] Figure 4: The ‘Assistant’ output box in panel (a) displays raw JSON data (visibility, keypoints_xyn) and a visualization, but the caption does not explicitly mention that the pipeline outputs structured pose data alongside the scene, which is a key component of the ‘subject-side guidance’.
- [writing] Figure 5: The caption states ‘Qualitative comparisons of different models,’ but the image contains two distinct sections: (a) a comparison of three models and (b) a demonstration of ShutterMuse’s interactive features. The caption should explicitly describe both parts to match the visual content.
- [writing] Figure 5: The sub-captions ‘(a)’ and ‘(b)’ are present in the image but are not referenced or described in the main figure caption, making the structure ambiguous.
- [science] Figure 6: The ‘Original Photo’ column displays images containing people (e.g., the person on the bicycle), which contradicts the caption’s description of ‘subject-side guidance’ where portrait images are converted into ‘person-free scenes’ to serve as the background for pose recommendations.
- [writing] Figure 6: The column headers (‘Original Photo’, ‘Nano-Banana-Pro’, etc.) are repeated for the top and bottom rows of images, creating visual clutter and redundancy.
- [science] Figure 7: The legend defines ‘Refine’ (yellow circle) and ‘Reject’ (green x) markers, but these series are not plotted in subplots (b) or (c), which only show ‘Keep’ and ‘Overall’. This creates ambiguity about whether ‘Refine’/‘Reject’ data is missing or intentionally omitted.
- [writing] Figure 7: Subplot (d) y-axis label is truncated to ‘num’ instead of ‘Number’ or ‘Count’, reducing clarity.
- [science] Figure 8: The caption states the second row shows ‘reject cases,’ but the image labels the second row as ‘Defects’ and the top row as ‘Refine’, ‘Reject’, ‘Keep’. The visual layout (3 columns) contradicts the caption’s description of rows (Refine, Reject, Keep). Specifically, the top row contains a ‘Reject’ example (middle column) and a ‘Keep’ example (right column), which conflicts with the caption claiming the first row is ‘refine

cases’.

- [writing] Figure 8: The legend at the top (‘Refine’, ‘Reject’, ‘Keep’) uses colored squares that do not match the background shading of the rows below. The top row has a pinkish background (matching ‘Refine’), the middle row has a blueish background (matching ‘Reject’), and the bottom row has a greenish background (matching ‘Keep’), but the text labels inside the rows (‘Defects’, ‘Strength’) do not align with the row’s assigned category in the caption.
- [science] Figure 9: The caption claims poses ‘float slightly,’ but the visualization shows severe structural failures where the skeleton is misaligned with the scene geometry (e.g., legs passing through the chair in the top row, torso floating in mid-air in the bottom row). The caption understates the magnitude of the failure shown.
- [writing] Figure 9: The figure lacks panel labels (e.g., a, b, c) to distinguish the different failure examples, making it difficult to reference specific cases in the text.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_jargon_police.md

The manuscript exhibits a high density of acronyms and specialized terminology that, while standard within the specific sub-field of multimodal AI and computer vision, may hinder accessibility for a broader audience or readers from adjacent disciplines. The primary issue is the introduction of acronyms without explicit definition at their first occurrence.

Specifically, the term MLLM (Multimodal Large Language Model) is used repeatedly in the Abstract and Introduction without being spelled out first. While the field is familiar with this term, the paper’s goal of “interactive assistants” suggests a broader potential audience that would benefit from the full expansion.

More critically, the authors introduce several custom acronyms for their proposed pipelines and metrics without definition. EMDP (expert-seeded, MLLM-verified self-distillation pipeline) and SGGP (subject-side guidance generation pipeline) appear in Section 3.1 and 3.2 respectively, but are never explicitly defined as acronyms before being used. This forces the reader to guess the meaning or search through the text to find the full phrase. Similarly, RFT (Reinforcement Fine-Tuning) is used in Section 5.1 before the full term is introduced in the same paragraph, creating a momentary confusion.

Standard evaluation metrics also suffer from this issue. BDE (boundary displacement error), RFR (ratio following rate), and SRCC (Spearman’s rank correlation coefficient) are introduced in Sections 3.2 and 6 without prior definition. While IoU is widely known, BDE and RFR are less common and should be defined. SRCC is a standard statistical term but should still be defined for completeness.

Additionally, SFT (Supervised Fine-Tuning) and GRPO (Group Relative Policy Optimization) are used in Section 4 without definition. vLLM is mentioned in Section 5.1 as a decoding accelerator without explanation. COCO is used in the context of keypoint formats; while it is a famous dataset, the specific “COCO 17-keypoint format” might benefit from a brief clarification for non-specialists.

To improve readability and inclusivity, the authors should ensure that every acronym is defined at its first use in the text. Custom pipeline names like EMDP and SGGP should either be defined immediately or replaced with descriptive phrases if they are not central to the paper’s core contribution. This will make the paper more accessible to a wider range of readers without sacrificing technical precision.

Action items.

- [writing] The manuscript exhibits a high density of acronyms and specialized terminology that, while standard within the specific sub-field of multimodal AI and computer vision, may hinder accessibility for a broader audience or readers from adjacent disciplines. The primary issue is the introduction of acronyms without explicit definition at their first occurrence. Specifically, the term MLLM (Multimodal Large Language Model) is used repeatedly in the Abstract and Introduction without being spelled out f

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_logical_consistency.md

The paper presents a coherent framework for capture-time photography guidance, but there are specific logical inconsistencies between the definitions of tasks, the training objectives, and the evaluation metrics that require clarification.

First, the definition and evaluation of the “reject” class in the photographer-side task lack logical precision. The Introduction and Dataset sections define “reject” as images with “non-croppable defects” (e.g., severe blur, tilt). However, the evaluation metric “Reject Success Rate (RSR)” is defined simply as the percentage of ground-truth reject samples correctly classified. While this is a standard classification metric, the logical link to the *utility* of the guidance is weak. If a model predicts “reject” for a blurry image, it is correct, but the paper does not explain how this decision is distinct from a “refine” decision that fails to find a valid box. More critically, the MLLM-Score definition for “reject” samples assigns a score of 1 if the model predicts “reject” and 0 otherwise. This creates a tautology where the score is entirely determined by the classification accuracy, ignoring the *reasoning* or the *quality* of the rejection decision (e.g., did the model correctly identify the specific defect?). The logical flow from “defect identification” to “score assignment” is circular and does not measure the model’s ability to distinguish *why* an image is unfixable versus fixable.

Second, there is a disconnect between the training reward and the evaluation ground

truth in the photographer-side refinement task. The RFT reward function R_{mask} (Eq. 10) optimizes for the preservation of salient objects detected by BiRefNet. However, the ground truth for “refine” samples is defined by expert-annotated composition boxes, which are based on aesthetic principles, not just salient object preservation. It is logically possible for an expert to prefer a crop that excludes a salient but distracting object, or to include a non-salient background element for context. By optimizing strictly for salient object coverage, the model is trained to maximize a proxy (salience) that may not correlate perfectly with the target (expert aesthetic composition). The paper does not provide evidence or a logical argument that maximizing salient object coverage is a sufficient condition for achieving high IoU with expert boxes. This misalignment between the training signal (salience) and the evaluation metric (expert box overlap) weakens the causal claim that the RFT stage improves aesthetic composition.

Third, the subject-side guidance task exhibits a contradiction between the training objective and the evaluation premise. The evaluation section states that “multiple poses may be plausible for the same scene” and that reference keypoints are used to “characterize plausible pose configurations” rather than as a single target. However, the training data construction and the RFT reward function R_{sub} (Eq. 12) enforce an exact match between the predicted and ground-truth visibility vectors ($\mathbf{v}_{\text{pred}} = \mathbf{v}_{\text{gt}}$). This creates a logical inconsistency: the model is trained to converge on a single, specific visibility state (the ground truth), while the evaluation claims to reward a broader set of plausible poses. If the ground truth visibility is just one of many valid options, training the model to match it exactly may penalize it for generating other valid poses that differ in visibility (e.g., a keypoint being occluded vs. visible). The training objective should logically reflect the evaluation’s acceptance of multiple valid solutions, perhaps by using a relaxed matching criterion or a reward that accounts for the plausibility of the predicted pose rather than exact visibility matching.

These issues do not invalidate the core contribution but require clarification to ensure the logical consistency of the methodology and the validity of the conclusions drawn from the experiments.

Action items.

- [science] Clarify the logical link between the ‘reject’ class definition (non-croppable defects) and the RSR/MLLM-Score metrics. The current scoring for ‘reject’ samples appears circular (score=1 iff predict ‘reject’), failing to measure the model’s ability to identify specific defects versus simply classifying.
- [science] Justify why optimizing for salient object coverage (R_{mask}) logically leads to expert-defined ‘refine’ outcomes. Experts may exclude salient but distracting objects, creating a misalignment between the RFT reward signal and the IoU/MLLM-Score evaluation targets.
- [science] Resolve the contradiction between training for exact visibility matching (R_{sub}) and evaluating for ‘multiple plausible poses’. Training on a single ground-truth visibility state conflicts with the evaluation premise that multiple valid solutions exist.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_overreach.md

The paper makes several strong claims regarding the performance and capabilities of ShutterMuse that slightly exceed the evidence provided in the results tables.

First, the Abstract and Introduction repeatedly claim that ShutterMuse achieves the “best overall photographer-side performance among evaluated baselines.” While ShutterMuse leads in IoU, BDE, and Refinement Success Rate (R), Table 1 clearly demonstrates that it does not outperform all baselines on every metric. Specifically, Qwen3-VL-32B achieves a Keep Success Rate (KSR) of 98.18%, significantly higher than ShutterMuse’s 74.55%. Similarly, Gemini-3.0-Pro achieves a higher Reject Success Rate (RSR) of 82.76% (tied with ShutterMuse) but a higher KSR. The claim of “best overall” implies a dominance that is not present in the data. The authors should refine this claim to “best overall balance” or explicitly state that it leads in refinement accuracy while maintaining competitive decision-making, rather than implying a blanket superiority.

Second, the claim of “competitive subject-side pose recommendation” in the Abstract and Introduction is somewhat generous given the quantitative results in Table 2. ShutterMuse achieves a mean score of 0.34, which is lower than both Nano-Banana-Pro (0.39) and GPT-Image-2 (0.35). While the inference time is substantially lower (4.96s vs 55.16s/102.61s), the quality gap is noticeable. The text currently frames this as “competitive,” which might mislead readers into thinking the quality is on par. A more accurate characterization would be that it “approaches the performance of larger foundation models with significantly lower inference cost” or “offers a favorable trade-off between quality and efficiency.”

Finally, the User Study section concludes that the MLLM-based evaluation “aligns well with human judgments.” This conclusion is drawn from a study involving only six participants and 100 samples per subset. While the reported Spearman’s rank correlation coefficient (SRCC) of 0.90 is high, extrapolating this to a general alignment with “human judgments” across the broad domain of photography is an overreach. The conclusion should be more conservative, acknowledging the limited scale of the user study and framing the alignment as observed within the specific experimental context.

These issues are primarily matters of precise wording and calibration of claims to the data, rather than fundamental flaws in the methodology.

Action items.

- [writing] The paper makes several strong claims regarding the performance and capabilities of ShutterMuse that slightly exceed the evidence provided in the results tables. First, the Abstract and Introduction repeatedly claim that ShutterMuse achieves the “best overall photographer-side performance among evaluated baselines.” While ShutterMuse leads in IoU, BDE, and Refinement Success Rate (R), Table 1 clearly demonstrates

that it does not outperform all baselines on every metric. Specifically, Qwen3-VL-3

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_safety_ethics.md

The paper presents a novel benchmark and model for capture-time photography guidance, which inherently involves interacting with human subjects and their environments. From a safety and ethics perspective, the primary concerns revolve around data privacy, consent, and the potential for generating harmful recommendations.

Data Privacy and Consent: The construction of the subject-side dataset (Section 4.1, “Subject-side guidance”) involves a critical step where portrait images are processed using “Nano-Banana-Pro” to remove the subject, creating a “person-free scene” (lines 235-238). The paper states that these images are collected from “multiple online platforms” (line 226) but does not specify the nature of these platforms or the consent status of the individuals depicted. Using images of people, even if the person is later removed, for training a model that generates pose recommendations raises significant privacy and ethical questions. The authors must explicitly disclose the source of these images (e.g., are they from public datasets with known licenses, or scraped from social media?) and confirm that appropriate consent was obtained for the use of these images in this research context. If the images were scraped, the ethical implications of using personal likenesses without explicit consent for training a generative model need to be addressed.

Safety of Generated Content: The model is designed to recommend human poses (Section 1, “Introduction”; Section 4.2, “Subject-side guidance”). There is a non-trivial risk that the model could recommend poses that are physically dangerous (e.g., balancing on unstable objects, extreme contortions) or culturally insensitive/inappropriate for the specific scene. While the paper mentions that “professional annotators subsequently review and revise these rationales” (line 248) and that “Five experienced photographers further verify both the generated rationales and the pose annotations” (line 253), it does not detail the specific safety criteria or guidelines provided to these annotators. The review should include a statement on whether safety constraints were applied during data generation or model training to prevent the suggestion of hazardous poses. Additionally, the “Failure Case Analysis” (Appendix D) notes issues with “floating” poses but does not mention any instances of unsafe pose recommendations, which might suggest a lack of rigorous safety testing or reporting.

Evaluation Ethics: The evaluation relies heavily on MLLM judges (Section 4.2, “Subject-side evaluation metrics”). While this reduces the need for human subjects in the evaluation phase, the training data generation involved human annotators. The paper should clarify if the annotators were trained to identify and reject any poses that could be interpreted as

unsafe or harmful. Furthermore, the “User Study” (Section 6) recruited six participants for blind evaluation. The paper should briefly mention the ethical approval process for this study (e.g., IRB approval or informed consent procedures), although this is often standard practice, it is good to explicitly state it in the context of human-subject research.

In summary, while the technical contributions are significant, the paper requires clarification on the ethical sourcing of the training data, particularly regarding the use of images containing human subjects, and a discussion on the safety measures implemented to prevent the generation of harmful pose recommendations.

Action items.

- [science] The dataset construction pipeline (Sec 4.1) relies on ‘Nano-Banana-Pro’ to remove subjects from portrait images to create ‘person-free scenes.’ This raises significant privacy and consent concerns regarding the source of the original portrait images. The paper must explicitly state the provenance of these images (e.g., public datasets, user-uploaded content) and confirm that appropriate consent was obtained for the original subjects before their likenesses were processed and used for training.
- [writing] The subject-side guidance task involves generating pose recommendations for humans in specific scenes. There is a risk of generating poses that could be unsafe, physically impossible, or culturally inappropriate for the specific context. The paper should include a discussion on safety filters or constraints implemented to prevent the model from suggesting harmful or dangerous poses, and clarify if any safety guidelines were provided to the human annotators during the verification process.
- [writing] The evaluation of subject-side guidance uses an MLLM judge to score ‘physical plausibility’ and ‘scene interaction’ (Sec 4.2, App B). While this avoids direct human testing, the training data generation involved human annotators reviewing poses. The paper should clarify if the annotators were provided with safety guidelines to reject poses that might lead to injury or if the dataset includes any exclusion criteria for hazardous scenarios.

Scientific Evidence

Weights the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_scientific_evidence.md

The paper presents a novel benchmark and model for capture-time photography guidance, but the scientific evidence supporting the quantitative claims requires further clarification regarding evaluation robustness and sample statistics.

Evaluation Reliability and Subjectivity: The subject-side guidance evaluation (Section 4.2, Table 2) relies exclusively on an MLLM judge (Gemini-3.0-Pro) to score “physical plausibility,” “scene interaction,” and “pose aesthetics.” While the authors report a high

correlation (SRCC=0.90) with human rankings in a small user study (n=100 samples, 6 participants), this does not fully validate the absolute scores or the fine-grained differences observed (e.g., ShutterMuse 0.34 vs. GPT-Image-2 0.35). The user study only validates *ranking*, not the *magnitude* of the scores. Given the high subjectivity of “aesthetics,” the paper lacks a report on inter-annotator agreement (IAA) for the human study or a larger-scale human evaluation to confirm that the MLLM judge’s absolute scoring is reliable. Without this, the claim of “competitive” performance based on a 0.01 difference in mean score is statistically fragile.

Sample Size and Class Imbalance: In Section 4.2, the authors state the photographer-side benchmark contains 421 held-out samples. However, the text does not explicitly break down the number of samples for the three decision classes: *refine*, *keep*, and *reject*. The results in Table 1 show specialized models achieving 0.00% RSR and KSR, while ShutterMuse achieves 82.76% and 74.55%. If the *keep* or *reject* classes are very small (e.g., <20 samples), these high percentages could be artifacts of small sample sizes rather than robust generalization. The authors must provide the exact class distribution (n_keep, n_reject, n_refine) to allow for a proper assessment of statistical significance and to rule out the possibility that the results are driven by class imbalance.

Proxy Metrics in Reinforcement Learning: The reinforcement fine-tuning (RFT) stage utilizes a reward function R_{mask} (Eq. 5) based on the coverage of a salient object mask generated by BiRefNet. While BiRefNet is a strong model, it is a general-purpose saliency detector, not a ground-truth annotation of “photographic composition.” Using it as a proxy for “correct composition” introduces a potential bias: the model may learn to optimize for what BiRefNet considers salient (often people or high-contrast objects) rather than what constitutes a good photograph in diverse contexts (e.g., negative space, abstract composition). The paper should discuss this limitation and how the reliance on an off-the-shelf saliency model might constrain the diversity of the learned guidance strategies.

Conclusion: The core methodology is sound, but the evidence for the subject-side scores and the robustness of the photographer-side decision metrics needs strengthening. Providing class counts and expanding the human evaluation analysis will significantly improve the scientific rigor of the claims.

Action items.

- [science] The subject-side evaluation relies entirely on MLLM judges (Gemini-3.0-Pro) for physical plausibility and aesthetics without reporting inter-annotator agreement or a robust human baseline beyond a small ranking study. Given the subjective nature of ‘aesthetics’ and ‘plausibility,’ the paper must clarify the reliability of the MLLM judge or provide a larger, statistically powered human evaluation to validate the 0.34 vs 0.35 mean score difference.
- [science] The photographer-side ‘Reject’ and ‘Keep’ success rates (RSR/KSR) show extreme variance across baselines (e.g., 0.00% for specialized models vs 82.76% for ShutterMuse). The benchmark size for these specific classes is not explicitly stated in

the text (only 421 total samples). Please report the exact class distribution (`n_keep`, `n_reject`, `n_refine`) to assess if the high RSR/KSR is driven by a small sample size or class imbalance.

- [science] The reinforcement learning reward $\$R_{\text{mask}}\$$ uses BiRefNet to define ‘salient object’ coverage. BiRefNet is a general-purpose model, not a photography-specific ground truth. The paper should discuss the potential noise introduced by using an off-the-shelf saliency detector as a proxy for ‘correct composition’ and how this might bias the model toward specific types of subjects (e.g., people) over others.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical analysis in this paper is generally sound in its design of metrics (IoU, BDE, success rates) but lacks rigorous reporting of uncertainty and significance testing, which is critical for claims of “best overall performance” and “high consistency” with human judgment.

1. Lack of Uncertainty Quantification in User Study: In Section 6 (“User Study”), the authors claim that the MLLM-based ranking is “highly consistent” with human preferences, citing a Spearman rank correlation coefficient (SRCC) of 0.90. However, the manuscript fails to report the statistical significance of this correlation (p-value) or a confidence interval. With a sample size of 100 examples and only 6 human participants, the variance of the SRCC estimate is non-negligible. Without a p-value or 95% confidence interval, it is impossible to determine if the observed correlation is statistically distinguishable from random chance or if the “high consistency” claim is robust. The authors should compute and report the 95% CI for the SRCC (e.g., via bootstrapping) and the associated p-value.

2. Single-Judge Reliability for MLLM-Score: The primary evaluation metric, MLLM-Score, relies on a single instance of Gemini-3.0-Pro as a judge. In Section 5.2, the authors state that each sample is assigned a score of $\{0, 0.5, 1\}$. There is no mention of inter-rater reliability (e.g., Krippendorff’s alpha) or intra-rater stability (e.g., running the judge multiple times with different seeds). Given that LLM judges can exhibit stochasticity, especially in subjective tasks like “aesthetics” or “plausibility,” reporting a single point estimate without error bars or variance is insufficient. The authors should report the standard deviation of the MLLM-Score across multiple independent judge runs or use an ensemble of judges to provide a more robust estimate of the metric’s reliability.

3. Statistical Significance of Ablation Results: Table 4 presents an ablation study comparing ShutterMuse-RL (Ours) against variants (e.g., w/o $\$R_{\text{dec}}\$$, w/o $\$R_{\text{mask}}\$$). The differences in metrics are often marginal (e.g., IoU 74.30% vs. 74.10%; RSR 82.76% vs. 79.31%). The text claims these components “contribute to the final performance,” but

it does not provide statistical evidence that these improvements are significant rather than noise. Given the fixed benchmark size (421 photographer-side samples), the authors should perform paired statistical tests (e.g., paired t-test or Wilcoxon signed-rank test) or bootstrap resampling to determine if the performance gains from the full model are statistically significant compared to the ablated versions.

4. Multiple Comparisons in Baseline Evaluation: In Table 1, the authors compare ShutterMuse against a large number of baselines (over 15 models). While the paper highlights ShutterMuse as the “best,” it does not address the issue of multiple comparisons. If the authors are making claims about ShutterMuse being significantly better than *all* baselines, a correction for multiple testing (e.g., Bonferroni or False Discovery Rate) should be considered, or at least a discussion on the risk of Type I errors in selecting the “best” model from a large pool.

In summary, while the experimental design is appropriate, the statistical reporting is incomplete. The paper requires the addition of confidence intervals, p-values, and significance tests to substantiate its claims of superiority and consistency.

Action items.

- [science] The user study (Sec. 6) reports a Spearman rank correlation of 0.90 between MLLM and human rankings but provides no confidence interval or p-value. With only 100 samples and 6 raters, the statistical significance of this correlation is unknown. Please report the 95% CI and p-value for the SRCC.
- [science] The MLLM-Score metric relies on a single judge (Gemini-3.0-Pro) without reporting inter-rater reliability or variance estimates. For the subject-side task, where multiple poses are plausible, a single deterministic score may be unstable. Please report the standard deviation of scores across multiple judge runs or multiple judges.
- [science] In the ablation study (Table 4), performance differences between variants (e.g., IoU 74.30 vs 74.10) are small. The paper does not state whether these differences are statistically significant or if they fall within the margin of error of the benchmark evaluation. Please add significance testing (e.g., paired t-test or bootstrap) for ablation comparisons.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_writing_quality.md

The manuscript demonstrates a high standard of academic writing, with a clear logical flow from the problem statement to the proposed solution and evaluation. The abstract effectively summarizes the contributions, and the introduction successfully motivates the need for capture-time guidance. The distinction between photographer-side and subject-side

tasks is articulated clearly throughout the text.

However, there are specific areas where precision and consistency could be improved to enhance readability and reproducibility. In Section 5, the description of the JSON output schema for the “reject” decision is ambiguous. The text states that an “empty value” indicates rejection, but does not specify whether this corresponds to a null field, an empty string, or a specific token in the JSON structure. Given the structured nature of the task, clarifying the exact JSON representation for all three decision classes (refine, keep, reject) is essential for readers attempting to replicate the system.

Additionally, Section 4.2 contains a commented-out paragraph and equation regarding the “ratio following rate (RFR)” metric. While the text explains the metric, its presence in a commented state within the final manuscript is confusing. If this metric was not used in the final evaluation, the commented code should be removed to prevent reader distraction. If it was used, the text should be uncommented and integrated into the results discussion.

Finally, there are minor inconsistencies in numerical precision. The abstract uses “approximately 130K samples,” while Section 4.1 provides exact counts (100K and 30K). While mathematically consistent, standardizing the phrasing to “130,000” or “130K” throughout would improve the professional polish of the paper. The definition of the coverage threshold τ_m is also deferred to the experimental setup section; moving this definition to the method section where the reward function is introduced would improve the self-containment of the methodology description.

Overall, the paper is well-written and the scientific narrative is strong. Addressing these specific points of ambiguity and formatting will further elevate the quality of the manuscript.

Action items.

- [writing] In Section 5 (ShutterMuse), the JSON schema description for photographer-side guidance states that an empty value indicates ‘reject’, but the text later says ‘[0,0,1,1] indicates keep’. This creates ambiguity: is ‘reject’ an empty string, null, or a specific token? Clarify the exact JSON structure for all three decision types to ensure reproducibility.
- [writing] Section 4.2 (CaptureGuide-Bench) introduces the ‘ratio following rate (RFR)’ metric with Equation 1, but the entire paragraph and equation are commented out in the LaTeX source. If RFR is not used in the final evaluation, remove the commented code to avoid confusion. If it is used, uncomment and ensure the text explains its inclusion in the results.
- [writing] The abstract and Introduction mention ‘130K samples’ for the dataset, but Section 4.1 specifies ‘100K photographer-side’ and ‘30K subject-side’. While the sum is correct, the phrasing ‘approximately 130K’ in the abstract followed by exact numbers later is slightly inconsistent. Consider using ‘130,000’ or ‘130K’ consistently to match the precision of the body text.
- [writing] In the ‘Reinforcement Fine-Tuning’ section, the text defines $R_{\text{ext}\{\text{mask}\}}$ and $R_{\text{ext}\{\text{photo}\}}$ but references a coverage threshold τ_m without defining it.

value in the text (it is defined in Section 6.1). For better flow, define $\text{\$ au_m\$}$ when first introduced in the method section or explicitly cross-reference the value.